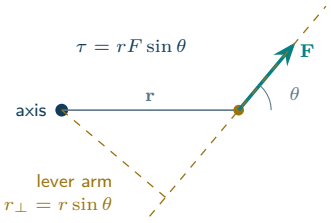
TORQUE ON ONE PAGE

The torque hinge

One rotational balance, $\sum \tau_{\text{ext}} = \mathrm{d}L/\mathrm{d}t$, and three origins of torque: gravity, friction and a magnetic couple.

1 / THE RULE AND THE PICTURE

Choose the axis first, and the lever arm decides how much of a force turns the body



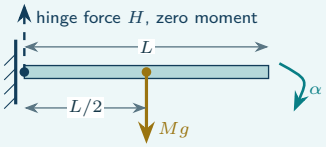
Begin with the chosen axis and the rotational balance, which for a rigid body on a fixed axis with constant I_{rot} becomes $\sum \tau = I_{\text{rot}}\alpha$; it links torque to angular acceleration and leaves the interaction unnamed.

$$\sum \tau_{\text{ext}} = \frac{\mathrm{d}L}{\mathrm{d}t}$$

Gravity: $\mathbf{r} \times M\mathbf{g}$. **Friction**: $\mathbf{r} \times \mathbf{f}$. **Magnetic**: a force couple.

2 / GRAVITY / TORQUE FROM A FIELD INTERACTION

A rod released from horizontal falls with $\alpha = 3g/2L$

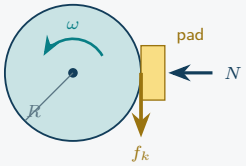


A uniform rod of 1.20 m and 3.0 kg, hinged at one end, is released from rest while horizontal. The weight acts at the centre of mass, $L/2$ from the hinge, so $\tau_g = MgL/2 = 17.6$ N m in the direction of fall, while the hinge force passes through the axis and has zero moment. With $I_{\text{rot}} = \frac{1}{3}ML^2 = 1.44$ kg m^2, $\alpha = 12.25$ rad s^{-2}, and the mass cancels because torque and inertia both scale with M, which does not make weight and inertia the same property. The dependence $\tau_g = Mg(L/2)\sin\theta$ belongs to the torque law and the geometry.

$$\tau_g = MgL/2 = 17.64 \text{ N m}, \qquad \alpha = 3g/(2L) = 12.25 \text{ rad s}^{-2}$$

3 / FRICTION / TORQUE FROM A CONTACT LAW

An empirical law and a geometric lever arm stop the flywheel

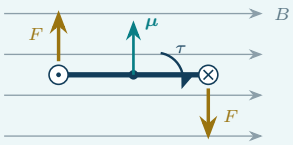


A brake pad presses on the rim of a 4.0 kg, 0.25 m solid disc turning at 18 rad s^{-1}, with $N = 10$ N and $\mu_k = 0.30$. The empirical law $f_k = \mu_k N = 3.0$ N and the geometry $\tau_f = f_k R = 0.75$ N m give a torque opposing ω; with $I_{\text{rot}} = \frac{1}{2}MR^2 = 0.125$ kg m^2, $\alpha = -6.0$ rad s^{-2} and $t_{\text{stop}} = 3.0$ s, confirmed by angular impulse ($-2.25 = I_{\text{rot}}\Delta\omega$ in kg m^2 s^{-1}). The 20.25 J of rotational energy becomes thermal, and the disc's L is not conserved because the brake exerts an external torque, though a system with brake, mounting and Earth can conserve it.

$$\tau_f = \mu_k NR = 0.75 \text{ N m}, \qquad t_{\text{stop}} = I_{\text{rot}}\omega/\tau_f = 3.0 \text{ s}$$

4 / MAGNETIC INTERACTION / A FORCE COUPLE

Equal and opposite forces sum to zero and still turn the coil



A 50-turn coil of area 8.0×10^{-3} m^2 carries 0.60 A in 0.25 T with its moment perpendicular to the field. The opposite side forces sum to zero but have separated lines of action, a couple, giving $\tau_B = NIAB\sin\theta = 6.0 \times 10^{-2}$ N m, where N, I, A, B set the strength and $\sin\theta$ the orientation (zero at $\theta = 0$). With $I_{\text{rot}} = 2.0 \times 10^{-2}$ kg m^2, $\alpha = 3.0$ rad s^{-2}; reversing the current reverses τ, doubling B doubles α and doubling I_{rot} halves it. The value is instantaneous, since θ changes and a real motor adds commutation, back emf and friction.

$$\tau_B = NIAB = 6.0 \times 10^{-2} \text{ N m}, \qquad \alpha = \tau_B/I_{\text{rot}} = 3.0 \text{ rad s}^{-2}$$

5 / SYNTHESIS

The symbol τ records the moment of an interaction without identifying it

case	axis	physical torque law	coupling or parameter	what I_{rot} represents
gravity	hinge	$\mathbf{r} \times M\mathbf{g} = Mg\frac{L}{2}\sin\theta$	g, centre-of-mass geometry	rod about the hinge, $\frac{1}{3}ML^2$
friction	axle	$\mathbf{r} \times \mathbf{f}$, with $f_k = \mu_k N$	μ_k, an empirical closure	disc about the axle, $\frac{1}{2}MR^2$
magnetic	coil axis	couple, $NIAB\sin\theta$	N, I, A, B	coil and rotor about the axis

$\sum \tau = I_{\text{rot}}\alpha$ needs a rigid body, a fixed axis and constant I_{rot}, so it is a specialised bridge from $\sum \tau_{\text{ext}} = \mathrm{d}L/\mathrm{d}t$.

ASIDE / NOTATIONAL COMPRESSION

The letter I is current in $NIAB$ and moment of inertia in $I_{\text{rot}}\alpha$. Torque and I_{rot} are not properties of a body alone: changing the axis changes the lever arm and I_{rot} together.

ASIDE / CHECK THE NUMBERS IN PYTHON

`3*9.8*0.6/(3*1.2**2/3)` gives 12.25, `-0.3*10*0.25/(0.5*4*0.25**2)` gives -6.0, and `50*0.6*8e-3*0.25/2e-2` gives 3.0.

Convention. After the worksheet *The Torque Hinge*, a companion to *The Placeholder Force*; see also *The Momentum Hinge*.

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